



The 50L Home Pilot

A Blueprint for Scaling Household Water Efficiency



KOHLER®



Speakers

About Us



Neil Patel

Manager, 50L Home Coalition

patel@wbcsd.org



Samantha McLelland

Enterprise Program Manager

samantha.mclelland@kohler.com

PART ONE

Why 50L?



Premise

Reduce water use by making sustainability irresistible

We're not *asking* households to use less water.

We're supporting performance-driven design and behavior change strategies that make home routines simpler and more convenient – water savings naturally follow.



Who we are

50L Home: a public-private platform for innovation

Co-Chairs



Members



Partners



Secretariat



Collaboration

Private sector: design and innovation

Utilities: service delivery

Government: regulation

Households: use behaviors

OUR MISSION

Make 50-liter living the standard through innovation that improves quality of life.



Opportunity

Why indoor water efficiency, why now



Reliable & consistent

Indoor use is stable across seasons, climate, and weather. Indoor savings don't evaporate in a wet year.



Globally replicable

A family in Los Angeles, London, or Chicago needs roughly the same indoor water.



Energy co-benefits

Water heating is roughly a fifth of residential energy use. Every gallon of hot water saved cuts emissions and bills.

Reframe

Reducing demand = solution to local challenges



Growing water demand

As suburbs deplete groundwater and increasingly turn to Lake Michigan, we squeeze our remaining legal allocation

Per capita demand reduction frees up headroom for additional users as the region grows



Urban flooding

Even with TARP, localized flooding and basement sewer backups occur (*source*: Chicago Green Infrastructure Strategy 2026)

Less daily wastewater volume frees up capacity in combined sewers when storms hit



Affordability

NE Illinois water rates rose ~80% over the past decade. Lower-income Chicago households spend up to 10% of income on water (*source*: Metropolitan Planning Council)

Every gallon saved cuts utility bills



Energy-water nexus

Hot water accounts for 18% of home energy use – second largest driver of energy bills (*source*: Department of Energy)

Reduced hot water consumption lowers energy bills.

PART TWO

Kohler Co.

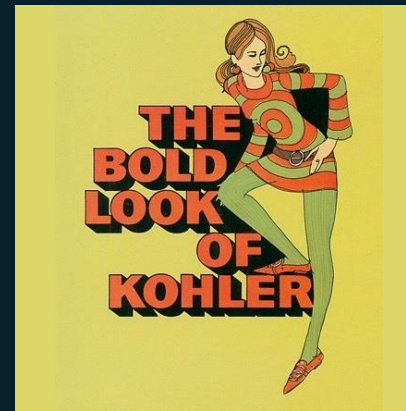
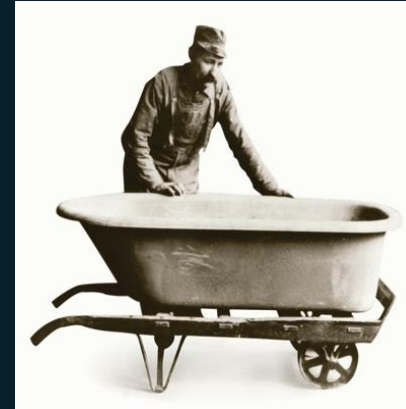
Kohler Company

OUR MISSION

Help people live gracious, healthy and sustainable lives.

Founded in 1873 in Sheboygan, WI. As a tiny machine shop and in 1883 made an innovative pivot to plumbing with Kohler's first bathtub.

More than a guiding principle, Sustainable Living, is our vision for the future—empowering customers and communities to make choices that enrich their lives while benefiting the planet. It's about striving to do better and making everyday decisions that together create lasting impact.



KOHLER

Help people live gracious, healthy and sustainable lives.

Water Stewardship Through Uncompromising Design

*Water is **essential** to life and core to **what we do at Kohler**; we are innovating to enable and preserve **access to clean and efficient** water experiences for all, now and for **generations to come**.*



PART TWO

What we've learned

Two years, 31 homes, fixture-level data, and a path to apply it in Chicago

LA Pilot

Two years, three phases, fixture-level data

31

households

15

retrofit / 16 control

2 yrs

of measurement

6

submeters per home

M E T H O D O L O G Y

- Whole-home Flume monitors plus Droople fixture submeters at six points per home
- Minute-by-minute flow rates and water temperatures
- Capture behavioral data on whole-home water use

PHASE 1 4 months

Baseline

Existing fixtures and appliances measured across all 32 homes.

PHASE 2 12 months

Market-available retrofits

Latest products you can buy today Kohler/IKEA fixtures, Electrolux W/D, P&G consumables.

PHASE 3 12 months

Prototype products

Next-generation fixtures (~30% below code), prototype appliances, recirculating shower test.

LA Pilot

High-frequency monitoring at the appliance level



Headline Results



21

**gallons / person /
day**



56%

**less water use than LA
household average**



4.4

/ 5 satisfaction

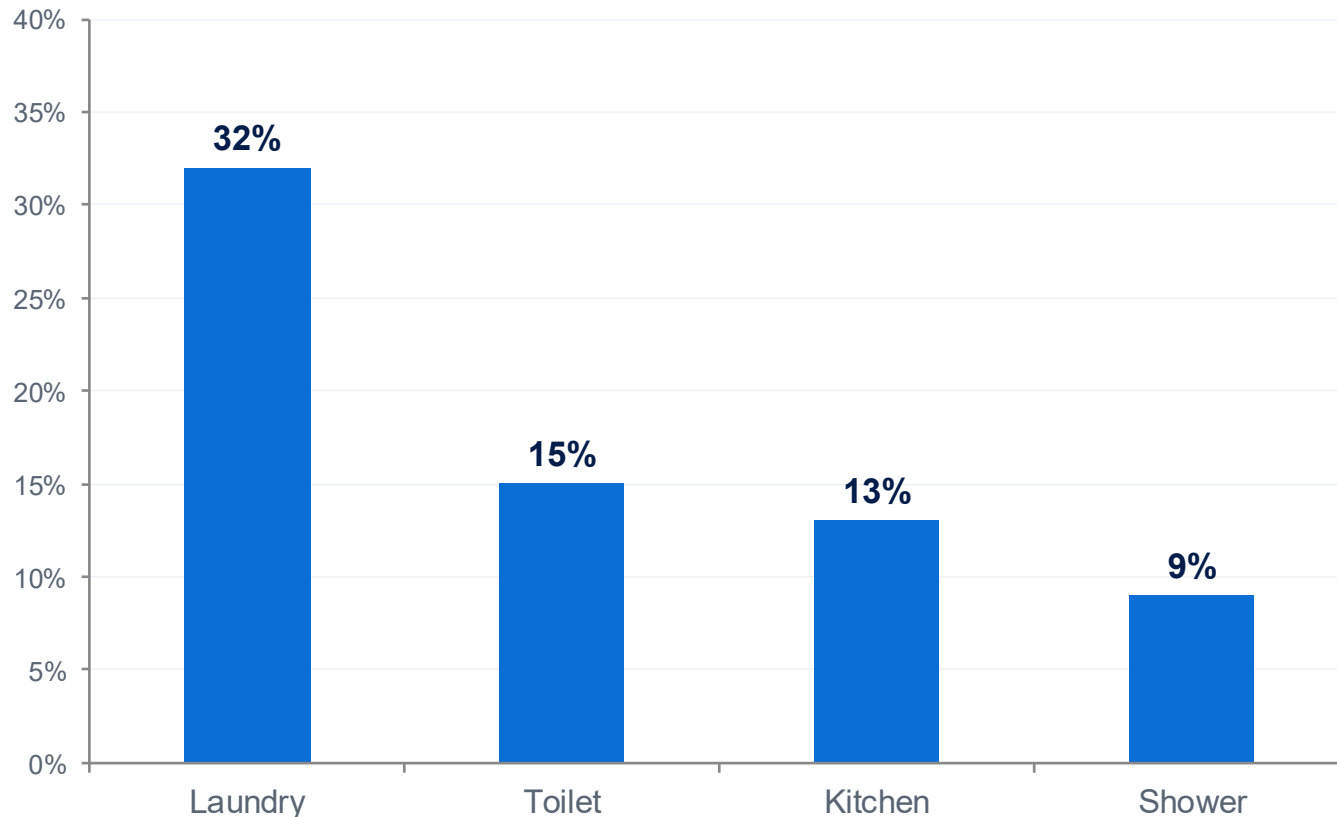


18%

**reduction in energy
use relative to
baseline**

By the Numbers

Reductions in water use by category

**CUSTOMER BENEFITS****Kitchen (HE Faucet/Dishwasher + Soaps)**

- ✓ No pre-rinsing
- ✓ Minimal soaking/scrubbing

Laundry (HEFL Washer + Coldwater Detergent)

- ✓ Fewer loads
- ✓ No sorting
- ✓ No hot water required – 18% energy savings

Shower (HP Showerhead + 2-in-1 Products)

- ✓ Kohler 1.25 GPM showerheads reduced streamflow with no behavioral compensation

Toilet (0.8 GPF Toilet + HP Toilet Paper)

- ✓ Reduced double flushing
- ✓ Fewer clogs

Insight 1

Pairing hardware and consumables enables behavior change

No single product delivered the full reduction. The savings came from pairings: efficient hardware plus high-performing consumables.



HE Dishwasher + Cascade + Dawn
Powerwash

44%

kitchen sink water reduction

Pre-rinsing essentially eliminated. The appliance enables the behavior change; the consumable removes the friction.



HE Washer + Tide Coldwater Clean

54%

laundry hot water reduction

Cold-water washing became default, not the exception. Sorting by color became unnecessary. Compounded water and energy savings.

Insight 2

Choice architecture matters



Dual options increase water use.

Dual-flush toilets actually increased water use by 35%.

Participants reported confidence in using them correctly but defaulted to the higher-volume button.

Rinse booster features increased water use by 24%



Participants stick to default options.

Even highly engaged participants left savings on the table:

18 of 26 never tried eco dishwasher cycles

24 of 26 never tried the dryer refresh feature

Additional savings achievable through education and prompt

Business Case

Positive ROI and durable savings

\$5–7K

retail retrofit cost

Whole-home: dishwasher, kitchen faucet, washer, dryer, toilet, showerhead, bathroom faucets, plus a year of consumables.

34,500

gallons saved/year

Per household, every year — roughly equivalent to removing 1,000 sq ft of irrigated turf.

\$890

\$/year

Cost savings per household, every year

BENEFITS RELATIVE TO OUTDOOR EFFICIENCY

**Stable reductions in demand**

Indoor savings don't fluctuate with rainfall or season.

**Equitable**

Reaches rentals and multi-family housing, customer segments outdoor programs miss entirely.

**Direct-install ready**

Fits inside existing utility rebate and CAP infrastructure.

Key takeaways

Key takeaways by stakeholder and future work

**WATER/ POWER UTILITIES****Indoor water efficiency delivers reliable water & energy savings.**

- How can utilities integrate top-line efficiency products into existing rebate and incentive structures?
- What incentive models optimize the ratio of program expenses to household uptake?

**HOUSEHOLDS****Making 50L feel “irresistible”**

- What use behaviors drive actual water savings and how can design choices influence them?
- What additional savings can be achieved through nudges and education?

**GOVERNMENTS****Making 50L the standard for housing**

- How can we raise efficiency standards without compromising user experience?
- What reforms and code/permit pathways are required to enable broader adoption of 50L Home technology?

**INDUSTRY****Performance drives market success.**

- How can we continue to innovate to close the final stretch from 79L to 50L/day?

Future Work

Applications for the Great Lakes region

**AFFORDABILITY****Affordability-aligned direct installs**

Embed efficiency retrofits into customer assistance program (CAP) eligibility. Targets households where inefficient appliances drive higher bills, addressing affordability at the source.

**ENERGY EFFICIENCY****Energy savings through avoided hot water use**

Bundle with existing energy efficiency rebate / direct install programs. Hot water reduction captures water-energy nexus value; both utilities benefit, neither bears full program cost.

**URBAN FLOODING****Include water efficiency in stormwater incentives**

Stack stormwater incentives with efficiency incentives durable indoor water demand reduction with behavioral nudge programs to free sewer capacity during storms.

**NEW BUILDS****Integrate 50L specs at construction.**

Work with developers to integrate 50L Home specs into new builds at the design stage – leading to lifetime cost savings.

Questions?



Learn more: <https://50lhome.org>

Annexes

LA Pilot

Cost-benefit analysis

Product	Payback	Water Savings	10-Year Dollar Savings
Laundry (Washer + Detergent)	~3 years	56,756 liters 14,990 gallons	\$1,516
Kitchen (Dishwasher + Products)	~2 years	66,446 liters 17,560 gallons	\$1,357
Combined Indoor Bundle (Kitchen + Laundry)	As early as 2 years	123,000 liters 32,490 gallons	\$2,873

LA Pilot

Appliances used in LA pilot

Phase 2		Phase 3	
Kitchen  Low-flow kitchen faucet 1.5 gpm  High efficiency dishwasher 3.3 gal/cycle 		Kitchen  Ultra efficient kitchen faucet 1.0 gpm  Dishwasher with Smart Boost and Lux Care 	
Bathroom  Low-flow showerhead 1.5 / 1.8 gpm  Low-flow bathroom faucet 1.0 / 1.2 gpm  High efficiency (1.0 gpf) or dual-flush toilet (1.1 / 1.6 gpf) 		Bathroom  Dual flush toilet 1.0/0.8 gpf  Ultra efficient showerhead 1.25 gpm  Shower booster (on/off button)  Faucet mister  Recirculating shower and autodiverter 	
Laundry Room  High Efficiency Washer and Dryer 		Laundry Room  Perfect Steam Washer and Dryer with LuxCare Plus Wash and SmartBoost 	